

HAT-P-36b

R-1878

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250604 · created 2026-08-03T11:31:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460830.7558 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.11 +/- 0.031
Transit depth (Rp/Rs)^2	1.21 +/- 0.68 [%]
Orbital Inclination (inc)	80.23 +/- 2.95
Airmass coefficient 1 (a1)	0.944 +/- 0.012
Airmass coefficient 2 (a2)	0.0342 +/- 0.007
Scatter in the residuals of the lightcurve fit is	1.23 %
Best Comparison Star	#3 - [282, 259]
Optimal Aperture	3.12
Optimal Annulus	16.88
Transit Duration (day)	0.065 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.11 $\pm$ 0.031 vs 0.126 (deviation 0.52 σ)
Duration fit vs published	0.065 d vs 0.0925 d (deviation 1.15 σ)
Mid-time O-C	-15.8 min
Depth SNR	3.5
Residual scatter	1.23 %

Light curve

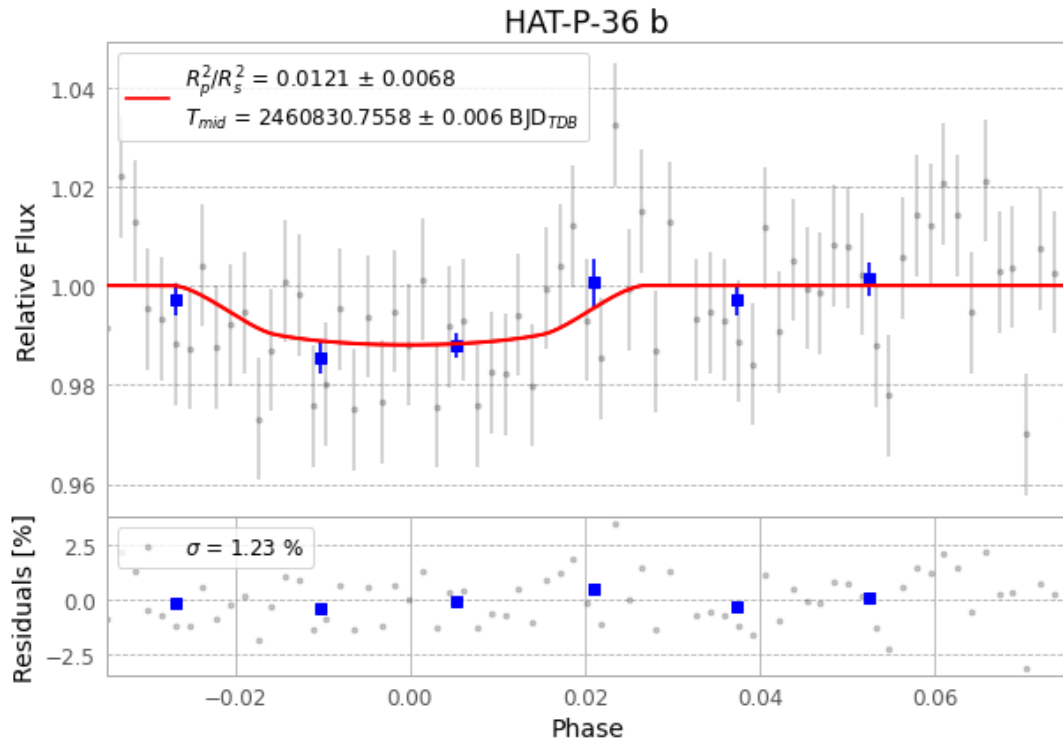


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-06-03.png (SHA-256 d9460291c7e9a54f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 98a24f3f33d32783...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), HATP-36250604050016.FITS → HATP-36250604083015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250604.log (3ffcde9dd671...), FinalLightCurve_HAT-P-36 b_2025-06-03.png (d9460291c7e9...), FinalLightCurve_HAT-P-36 b_2025-06-03.csv (cfd1b66e8aed...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 11:31Z.